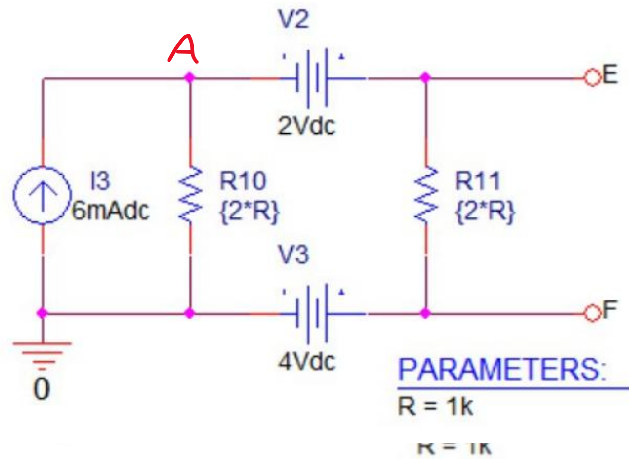


전자회로 해석 및 설계실무

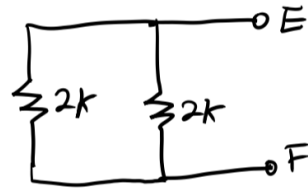
Homework #3

엄민호

1. E와 F 노드에서의 Thevenin / Norton 등가회로를 구하시오.



⇒ R_{Th} 구하기 (전류원 open, 전압원 short)



$$R_{Th} = 1k\Omega$$

$$V_{Th} = 5V, R_{Th} = 1k\Omega, I_N = 5mA$$

$$V_F = 0 + 4 = 4V, V_E = V_A + 2$$

노드 A에서 JCL

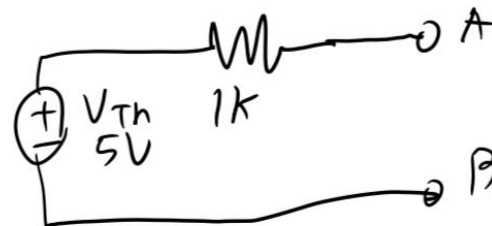
$$6m = \frac{V_A - 0}{2k} + \frac{V_A + 2 - 4}{2k}$$

$$\Rightarrow 12 = 2V_A - 2 \Rightarrow V_A = 7V$$

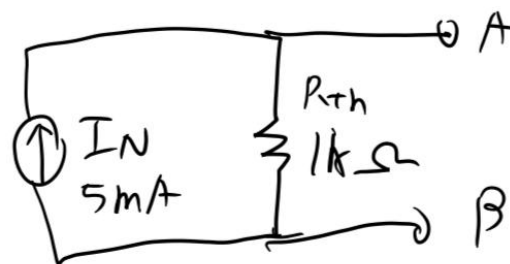
$$V_{Th} = V_E - 4 = 7 + 2 - 4 = 5V$$

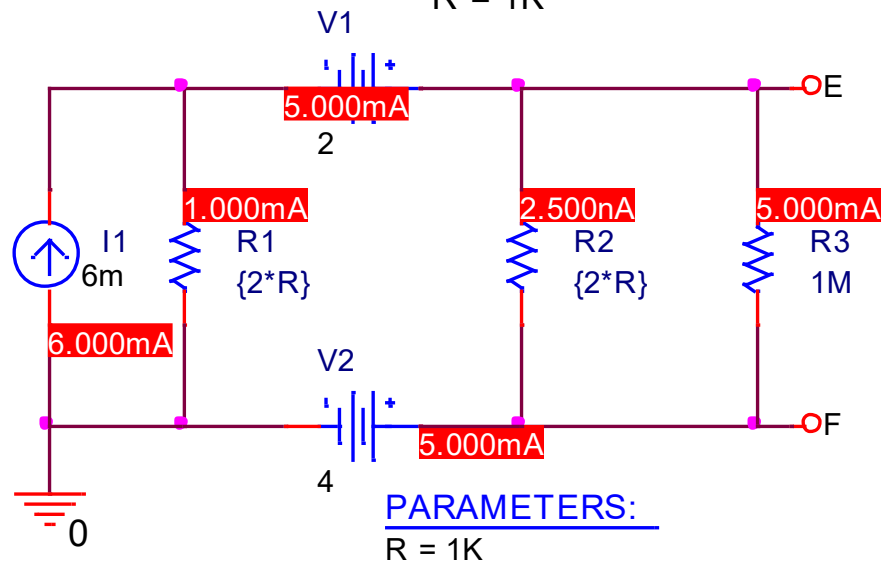
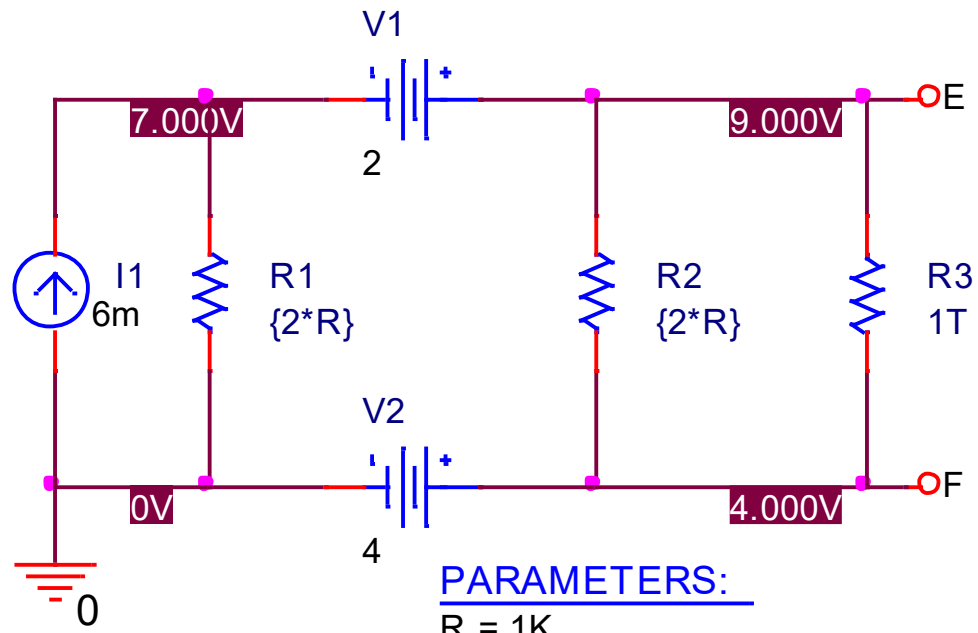
$$I_N = \frac{V_{Th}}{R_{Th}} = \frac{5}{1k} = 5mA$$

Thevenin 등가 회로

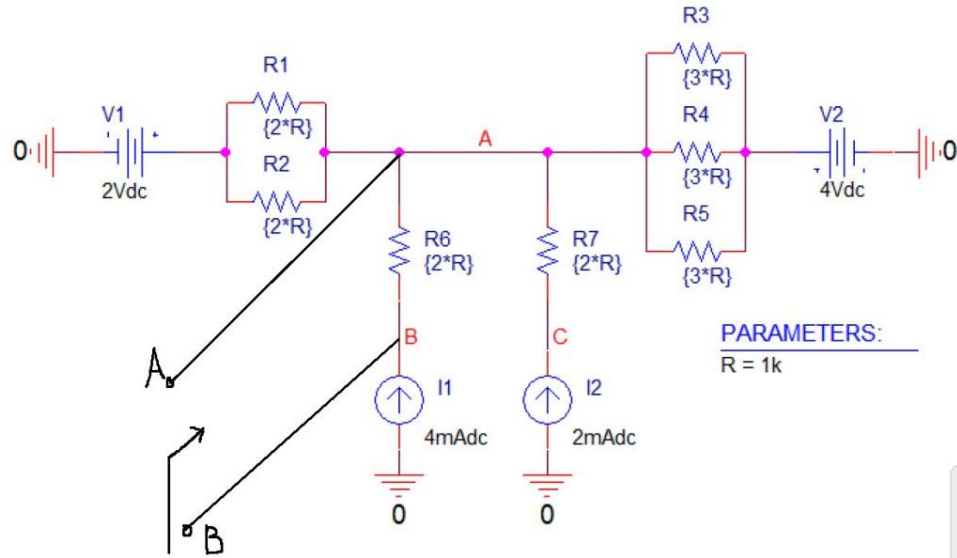


Norton 등가 회로





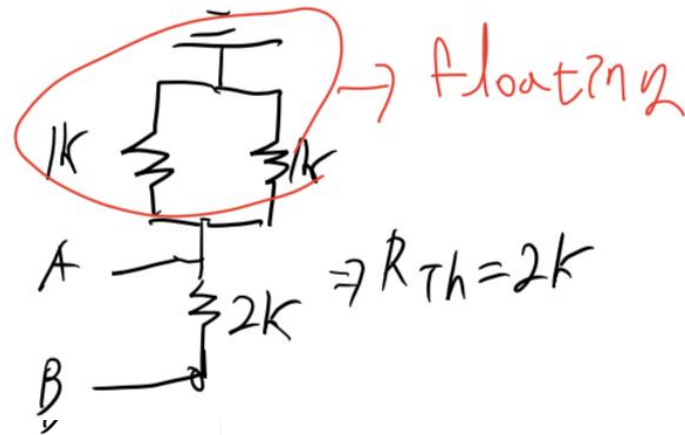
2. A와 B 노드에서의 Thevenin / Norton 등가회로 구하시오.



⇒ R_{Th} 구하기 (전압원 short, 전류원 open)

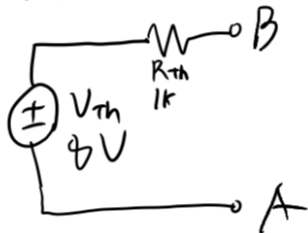
$$R1 // R2 \Rightarrow 2k // 2k = 1k \Omega$$

$$R3 // R4 // R5 \Rightarrow 3k // 3k // 3k = 1k \Omega$$



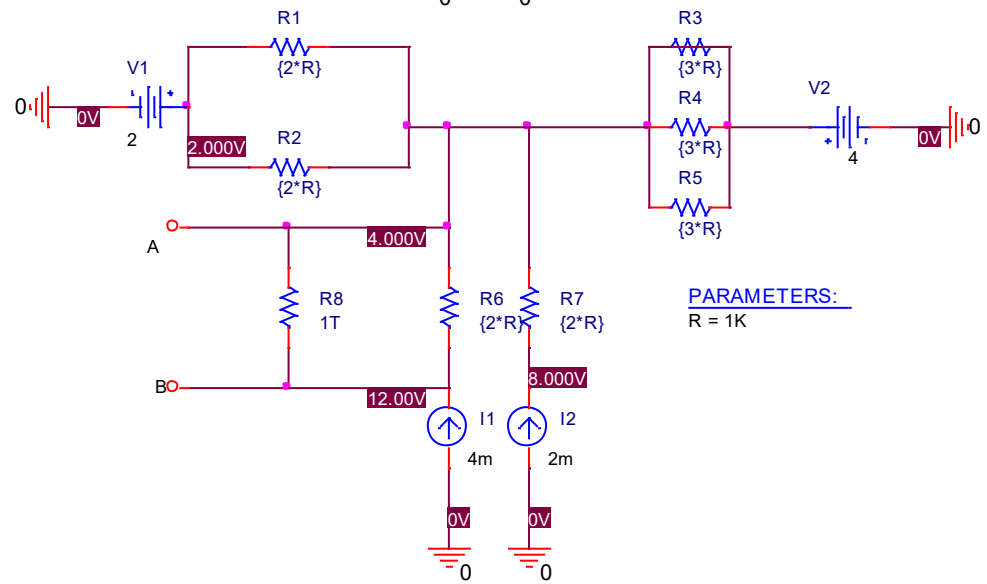
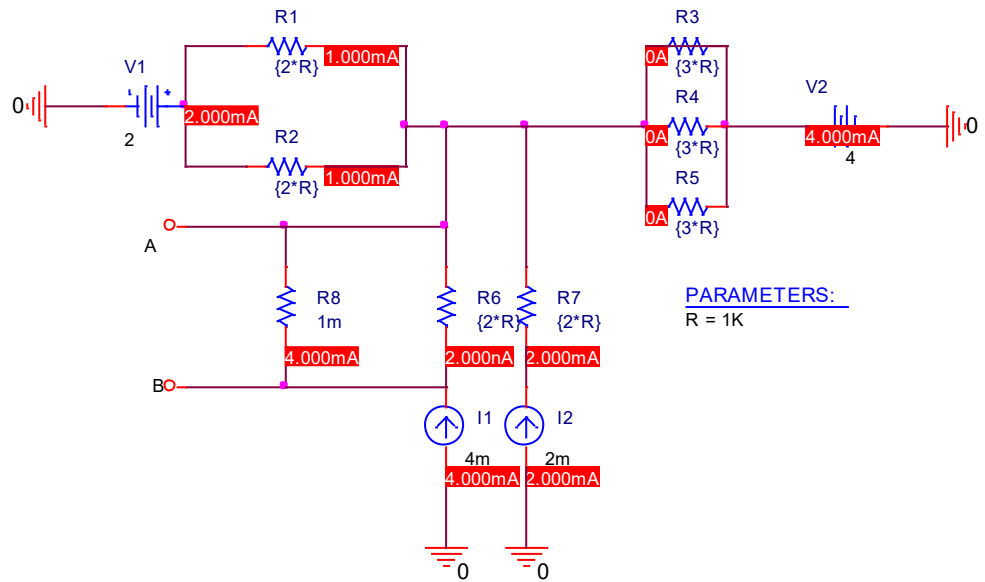
$$I_N = 4mA \Rightarrow V_{Th} = I_N \times R_{Th} = 4m \times 2k = 8V$$

Thevenin 등가회로

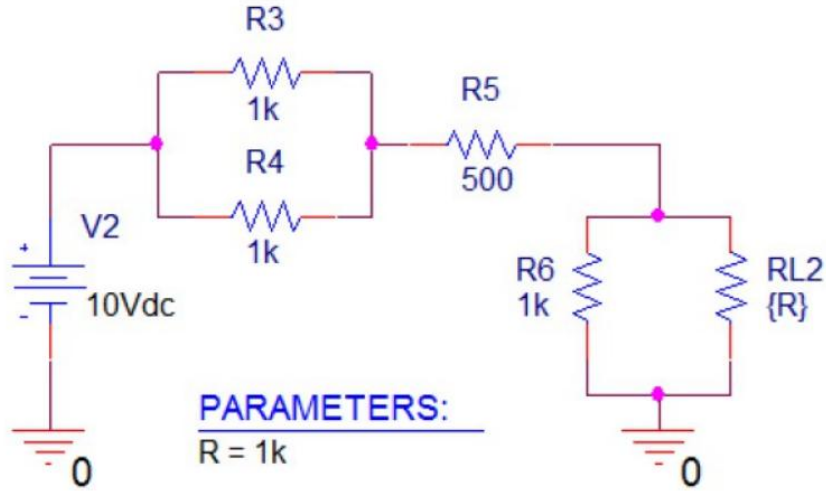


Norton 등가회로

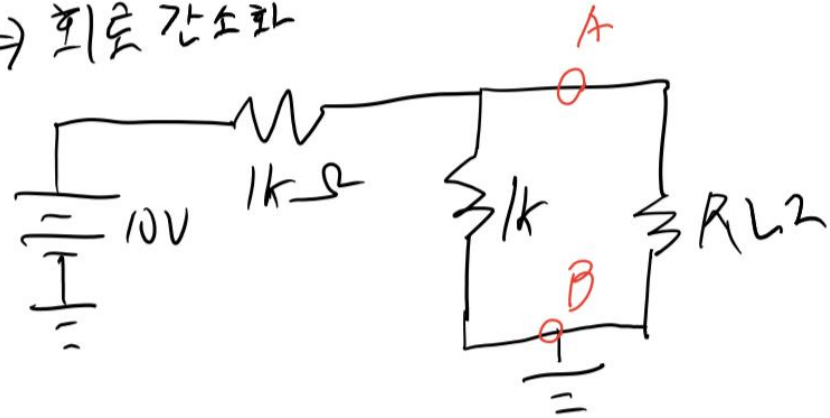




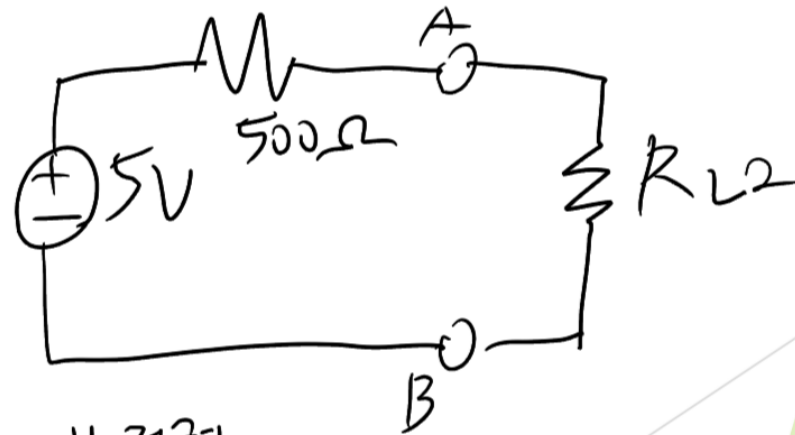
3. RL2 가 최대 전력을 소모하기 위한 저항값을 계산하시오.



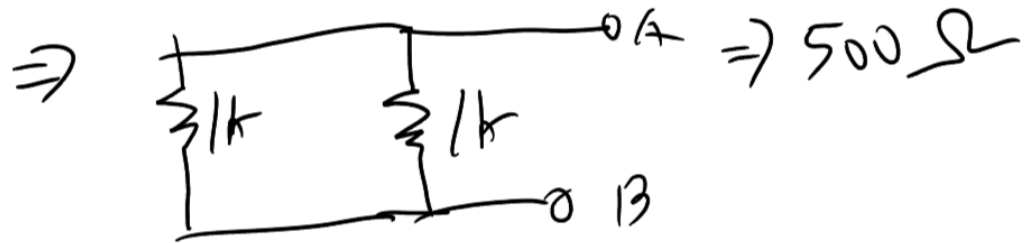
⇒ 회로 간소화



⇒ Thevenin 등가회로



노드 A, B 에서 들여다 본 저항



$$V_{Th} = 5V$$

⇒ $R_{L2} = 500\Omega$ 일 때 최대 전력

